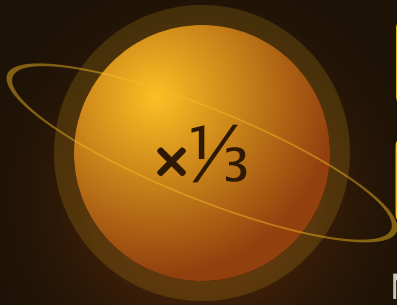




Fraction Products

Multiplying that shrinks, and dividing that grows.

THE BIG QUESTION

Why does dividing by $\frac{1}{2}$ give you more than you started with?

HANDS-ON PROJECT

Scale the Blueprint

Draw a room at half scale, then a quarter.
Every measurement multiplied by a fraction.

FLUENCY GAME

Bigger or Smaller

Call it before you compute. Will the answer be bigger or smaller than the number you started with?



AXIOM • MISSION COMMANDER, NINE TOURS

“How many halves fit in six? Twelve. That is all division by a fraction ever asks — and it is why the answer grows.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Write fractions as a/b . Items F1 and F2 want a prediction, not a calculation.

STRAND A
FRACTION OF A
WHOLE

A1 $1/2$ of 20

.....

A2 $3/4$ of 12

.....

STRAND B
FRACTION $\times$
FRACTION

B1 $1/2 \times 1/3$

.....

B2 $2/3 \times 3/4$

.....

STRAND C
MIXED NUMBERS

C1 $1\ 1/2 \times 4$

.....

C2 $2\ 1/2 \times 1\ 1/2$

.....

STRAND D
DIVIDING BY A
FRACTION

D1 $6 \div 1/2$

.....

D2 $1/2 \div 4$

.....

STRAND E
RECIPROCAL

E1 What times $1/4$ makes 1?

.....

E2 What times $2/3$ makes 1?

.....

STRAND F
PREDICTING SIZE

F1 Is $8 \times 3/4$ bigger or smaller than 8?

.....

F2 Is $8 \div 3/4$ bigger or smaller than 8?
Why?

.....

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Fraction of a whole	A1 A2	Week 1 • lesson 1.1	--- / 2	Day 1.1. This is Year One material.
B • Fraction × fraction	B1 B2	Week 1 • lessons 1.2 and 1.3	--- / 2	Day 1.2. Keep 1.3 — the area diagram is what makes Week 3 possible.
C • Mixed numbers	C1 C2	Week 2 • lessons 2.1 and 2.2	--- / 2	Compress Week 2 to three days.
D • Dividing by a fraction	D1 D2	Week 3 • all four lessons	--- / 2	Nothing without F2 also correct. A right answer here is often a memorised trick.
E • Reciprocals	E1 E2	Week 3 • lesson 3.2	--- / 2	Day 3.2. Move the flip-it practice into Warm-Up instead.
F • Predicting size	F1 F2	Week 1 • lesson 1.4 and Week 3 • lesson 3.1	--- / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

Read D1 and F2 together. A student who answers 12 to D1 but cannot say why the answer grew has been taught “keep, change, flip” and nothing else. That is worth undoing now, because in a year's time the same student will apply the flip to a multiplication and have no way of noticing.

A Fraction of a Whole

"Of" means multiply. Divide by the bottom, times by the top.

✦ Warm-Up

 $\frac{1}{2}$ of 20

 $\frac{1}{4}$ of 20

 $\frac{1}{3}$ of 12

 $\frac{1}{5}$ of 30

 $\frac{1}{2}$ of 50

 $\frac{1}{8}$ of 40

$\frac{3}{4}$ OF 12 • TWO STEPS, IN THIS ORDER

STEP 1 • $\div 4$

3

one quarter

STEP 2 • $\times 3$

9

three quarters

Divide by the bottom to find one part. Multiply by the top to find how many of those parts you want.

Doing it in the other order also works — $12 \times 3 = 36$, then $36 \div 4 = 9$ — but dividing first keeps the numbers small, which means fewer mistakes.

◆ Core

1. $\frac{3}{4}$ of 12
.....2. $\frac{2}{3}$ of 18
.....3. $\frac{5}{8}$ of 40
.....4. $\frac{3}{5}$ of 45
.....5. $\frac{2}{3}$ of 20. This one does not come out whole — write it as a mixed number.
.....

★ Challenge NOT GRADED

C1. $\frac{3}{4}$ of a number is 18. What is the number? Work backwards.

.....

C2. Which is more, $\frac{2}{3}$ of 30 or $\frac{3}{4}$ of 24? Answer before you calculate, then check.

.....



AXIOM: Divide by the bottom first. Smaller numbers, fewer mistakes, same answer.

Tops and Bottoms

Multiplying fractions is the easy one. Straight across, then simplify.

Warm-Up

$$\frac{1}{2} \times \frac{1}{2}$$

$$\frac{1}{2} \times \frac{1}{3}$$

$$\frac{1}{3} \times \frac{1}{3}$$

$$\frac{1}{4} \times \frac{1}{2}$$

$$\frac{2}{3} \times \frac{1}{2}$$

$$\frac{3}{4} \times \frac{1}{2}$$

WHY STRAIGHT ACROSS WORKS, AND WHY NO COMMON DENOMINATOR IS NEEDED

$$\frac{2}{3} \times \frac{3}{4}$$

$$\text{tops: } 2 \times 3 = 6$$

$$\text{bottoms: } 3 \times 4 = 12$$

$$\frac{6}{12} = \frac{1}{2}$$

Adding needs a common denominator because you are counting pieces, and the pieces must be the same size. Multiplying does not, because you are taking a part of a part — and the answer's pieces are new ones you create by cutting twice.

Two thirds of three quarters is a half. Notice the answer is smaller than both numbers you started with, which is what taking a part of a part always does.

Core SIMPLIFY EVERY ANSWER

1. $\frac{2}{3} \times \frac{3}{4}$

2. $\frac{3}{5} \times \frac{2}{3}$

3. $\frac{4}{5} \times \frac{5}{8}$

4. $\frac{3}{8} \times \frac{2}{9}$

5. In item 3, you could simplify before multiplying. Show how, and say which way was less work.

Challenge

C1. $\frac{2}{3} \times \frac{3}{2}$. Why does this one come out to exactly 1?

C2. Can multiplying two fractions ever give an answer bigger than both of them? Say when, or say why not.

Cut It Twice

The area picture. Fold one way, fold the other, count the overlap.

Warm-Up

HOW MANY PIECES AFTER TWO CUTS?

2×3

3×4

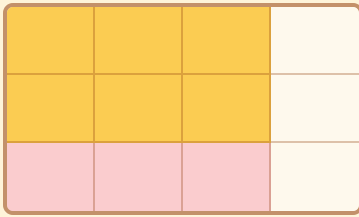
2×5

4×4

3×5

4×5

$2/3 \times 3/4$ • THE OVERLAP IS THE ANSWER



Gold = 6 of the 12 small boxes

Cut the square into 4 columns and shade 3 of them — that is $3/4$. Now cut it into 3 rows and take 2 of them — that is $2/3$.

The overlap is 6 boxes out of 12, which is $1/2$. The two cuts made $3 \times 4 = 12$ pieces, and the overlap is $2 \times 3 = 6$ of them. That is exactly why you multiply tops and multiply bottoms.

Core

DRAW THE GRID FOR EACH ONE

1. $1/2 \times 2/3$

.....

2. $3/4 \times 1/3$

.....

3. $2/5 \times 1/2$

.....

4. $3/4 \times 2/5$

.....

5. In every answer above, the result is smaller than both fractions. Explain in one sentence why that always happens.

.....

Challenge

C1. Draw the grid for $3/4 \times 4/3$. How many boxes are in the overlap, and what does that tell you?

.....

C2. Use a grid to show $1/2$ of $3/4$ and $3/4$ of $1/2$ give the same answer. Why must they?

.....

Bigger or Smaller

Call it before you compute. The factor tells you which way it goes.

✦ Warm-Up BIGGER OR SMALLER THAN 12?

$12 \times 1/2$

12×2

12×1

$12 \times 3/4$

$12 \times 5/4$

$12 \times 1/6$

ONE RULE, THREE CASES

FACTOR UNDER 1

Shrinks

$12 \times 3/4 = 9$. You are taking a part of it.

FACTOR OF EXACTLY 1

Stays

$12 \times 4/4 = 12$. One whole copy, unchanged.

FACTOR OVER 1

Grows

$12 \times 5/4 = 15$. More than one copy.

Compare the top to the bottom. Top smaller means the fraction is under one, so the answer shrinks. That check takes a second and catches most errors in this mission.

◆ Core PREDICT FIRST, THEN COMPUTE

1. $8 \times 3/4$
.....

2. $8 \times 5/4$
.....

3. $20 \times 2/5$
.....

4. $20 \times 7/5$
.....

5. Somebody says $20 \times 2/5 = 50$. Use the bigger-or-smaller rule to show that is wrong.
.....

★ Challenge

C1. What fraction would you multiply 12 by to get exactly 18? Find it, then say how you knew it had to be bigger than 1.
.....

C2. $100 \times 99/100$. Bigger or smaller than 100, and by how much?
.....

Bigger or Smaller

Call it out loud before either player computes anything.

How to play: one player writes a whole number and a fraction, and an operation — either $\times$ or $\div$. Both players call “bigger” or “smaller” before anyone calculates. Then work it out. A correct call scores a point; a correct call with a correct reason scores two.

✦ Warm-Up • call it, then check

$6 \times 1/2$ — bigger or smaller than 6?

.....

$6 \div 1/2$ — bigger or smaller than 6?

.....

Same two numbers, opposite answers. That pair is the whole mission in one line.

◆ Play eight rounds

ROUND	PROBLEM	MY CALL	ANSWER	POINTS
1				
2				
3				
4				
5				
6				
7				
8				

★ Challenge

C1. Write the rule for division in one sentence: dividing by a fraction under 1 does what, and why?

.....

C2. Find an operation and a fraction that leave 12 completely unchanged. How many can you find?

.....

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ • end of week 2

TEST • end of week 4

Week 2 • Mixed Numbers QUIZ FRIDAY

Convert to improper fractions first, every time. Trying to multiply mixed numbers directly is where this week goes wrong.

MON 2.1

Convert first

1 $\frac{1}{2}$ becomes $\frac{3}{2}$ before anything else happens.

TUE 2.2

Mixed $\times$ whole

Four batches of 1 $\frac{1}{2}$ cups.

WED 2.3

Mixed $\times$ mixed

2 $\frac{1}{2} \times 1 \frac{1}{2}$, and why it is not 2 $\frac{1}{4}$.

THU 2.4

Simplify early

Cancel before you multiply and the numbers stay small.

FRI

Mid-unit quiz

Page 10. Eight items, 85% to keep flying.

Week 3 • Dividing by a Fraction THE MISSION

Start with “how many fit inside” and the flip rule arrives as a shortcut he can explain, rather than a spell.

MON 3.1

How many fit inside

How many halves in six? Draw it before you write it.

TUE 3.2

Reciprocals

The number that multiplies with yours to make 1.

WED 3.3

Why flipping works

The shortcut, derived rather than announced.

THU 3.4

Fraction $\div$ fraction

$\frac{3}{4} \div \frac{1}{8}$, with a sanity check on the size.

FRI

Blueprint begins

Measure a real room, full size, on paper.

Week 4 • Proof UNIT TEST

The blueprint gets drawn at two scales, and the scaling arithmetic gets defended.

MON 4.1

Half scale

Every measurement $\times \frac{1}{2}$.

TUE 4.2

Quarter scale

Half of the half. Does the area halve too?

WED 4.3

Back to full size

Divide by the scale — and watch the number grow.

THU

Error journal sweep

Fix only what repeats.

FRI

Mission 06 test

Pages 11–12. Twelve items plus one explanation.

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. $\frac{3}{4}$ of 12

.....

2. $\frac{1}{2} \times \frac{1}{3}$

.....

3. $\frac{2}{3} \times \frac{3}{4}$

.....

4. $\frac{4}{5} \times \frac{5}{8}$

.....

5. $1\frac{1}{2} \times 4$

.....

6. $2\frac{1}{2} \times 1\frac{1}{2}$

.....

7. $20 \times \frac{2}{5}$

.....

8. $\frac{2}{3} \times \frac{3}{4}$ comes out smaller than both fractions. Explain why multiplying two fractions under 1 always does that.

.....

.....

Marking note: item 8 wants the idea of taking a part of a part. Accept any answer along the lines of "you are taking a piece of something that is already only a piece."

Unit Test • Part 1

Items 1–8. No time limit. Predict bigger or smaller before each one.

1. $\frac{2}{3}$ of 18

.....

2. $\frac{3}{5} \times \frac{2}{3}$

.....

3. $\frac{3}{8} \times \frac{2}{9}$

.....

4. $2\frac{1}{2} \times 1\frac{1}{2}$

.....

5. $6 \div \frac{1}{2}$

.....

6. $\frac{1}{2} \div 4$

.....

7. $\frac{3}{4} \div \frac{1}{8}$

.....

8. What times $\frac{2}{3}$ makes 1?

.....

Unit Test • Part 2

Items 9–12 and the Big Question.

9. $2\frac{1}{4} \div \frac{1}{2}$

.....

10. $100 \times 99/100$

.....

11. A room is 12 ft by 9 ft. Draw it at half scale — what are the two measurements? At quarter scale?

.....

12. A recipe calls for $\frac{3}{4}$ cup of oats per batch. You have 6 cups. How many batches can you make?

.....

EXPLAIN YOUR THINKING • THE BIG QUESTION

Why does dividing by $\frac{1}{2}$ give you more than you started with?

Use $6 \div \frac{1}{2}$ as your example, and say what the question is really asking. Scored on reasoning, not neatness.

.....
.....
.....
.....
.....

11–12

Gold band. The Scaling Badge is earned.

9–10

Silver band. One reteach day, then the badge.

≤ 8

Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 • 10 | A2 • 9 | B1 • $\frac{1}{6}$ | B2 • $\frac{1}{2}$

C1 • 6 | C2 • $3\frac{3}{4}$ | D1 • 12 | D2 • $\frac{1}{8}$

E1 • 4 | E2 • $\frac{3}{2}$ | F1 • smaller | F2 • bigger

D1 with F2 wrong is the pattern to watch — see the skip chart note. A student who says “bigger, because you're asking how many three-quarters fit in 8” has the whole mission already.

Lesson 1.1 • A Fraction of a Whole

Warm-Up • 10 | 5 | 4 | 6 | 25 | 5

Core • 1 • 9 | 2 • 12 | 3 • 25 | 4 • 27 | 5 • $13\frac{1}{3}$

C1 • 24 — one quarter is 6, so the whole is 24. C2 • both are 20 and 18, so $\frac{2}{3}$ of 30 is more; the point is that the prediction is genuinely hard here, which makes calculating worthwhile.

Lesson 1.2 • Tops and Bottoms

Warm-Up • $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{9}$ | $\frac{1}{8}$ | $\frac{1}{3}$ | $\frac{3}{8}$

Core • 1 • $\frac{1}{2}$ | 2 • $\frac{2}{5}$ | 3 • $\frac{1}{2}$ | 4 • $\frac{1}{12}$

Item 5 • cancel the 5s first, leaving $\frac{4}{1} \times \frac{1}{8} = \frac{1}{2}$ — far less work than $\frac{20}{40}$. C1 • because $\frac{2}{3}$ and $\frac{3}{2}$ are reciprocals. C2 • only if at least one fraction is greater than 1; two proper fractions always give something smaller than both.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 • Cut It Twice

Warm-Up • 6 | 12 | 10 | 16 | 15 | 20

Core • 1 • $1/3$ | 2 • $1/4$ | 3 • $1/5$ | 4 • $3/10$

Item 5 • because you are taking only a part of something that was already only a part. C1 • 12 boxes out of 12, so the answer is 1 — the grid makes reciprocals visible. C2 • both give $3/8$; the grid is the same square either way, just described in the other order.

Lesson 1.4 • Bigger or Smaller

Warm-Up • smaller | bigger | same | smaller | bigger | smaller

Core • 1 • 6 | 2 • 10 | 3 • 8 | 4 • 28

Item 5 • $2/5$ is less than 1, so the answer must be under 20; 50 is bigger, so it cannot be right. C1 • $3/2$, and it had to exceed 1 because 18 is more than 12. C2 • 99, one less than 100 — worth noticing that multiplying by something just under 1 barely changes anything.

Friday • Bigger or Smaller

Warm-Up • $6 \times 1/2 = 3$, smaller | $6 \div 1/2 = 12$, bigger

C1 • dividing by a fraction under 1 makes the answer bigger, because you are asking how many small pieces fit inside — and small pieces mean lots of them. C2 • $\times 1$ and $\div 1$, in any disguise: $\times 4/4$, $\div 7/7$, and so on. Infinitely many, which is a nice thing to land on.

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

- 1 • 9
- 2 • $1/6$
- 3 • $1/2$
- 4 • $1/2$
- 5 • 6
- 6 • $3 \frac{3}{4}$
- 7 • 8
- 8 • part of a part

Unit test

- 1 • 12
- 2 • $2/5$
- 3 • $1/12$
- 4 • $3 \frac{3}{4}$
- 5 • 12
- 6 • $1/8$
- 7 • 6
- 8 • $3/2$
- 9 • $4 \frac{1}{2}$
- 10 • 99

Items 11, 12 and the Big Question

11 • half scale gives 6 ft by 4.5 ft; quarter scale gives 3 ft by 2.25 ft. Worth asking whether the area quartered — it did not, it went down to a sixteenth. **12** • 8 batches, since $6 \div 3/4 = 8$. **Big Question** • full marks for an answer that reframes the division as “how many halves fit inside six” and notices that small pieces mean lots of them fit. An answer that only says “because you flip it and multiply” scores half: the flip is the shortcut, not the reason, and a student who has only the shortcut has nothing to fall back on when they misremember it.

Error journal • Mission 06

ONE LINE PER MISTAKE • STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY WHAT I GOT WRONG, AND WHY
